

The Structure of Unscientific Revolutions

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In the view of many linguists working in syntax Chomsky's (1995), (1998), and (1999) minimalist program (MP) constitutes a major paradigm change in the theory of grammar. On the Government Binding (GB) model the well-formedness of a sentence is determined by a set of conditions which apply locally to each element in the sequence of structures that comprises its derivation. Therefore, GB represented a move towards the constraint-based approach that characterizes many non-derivational theories, such as HPSG and LFG. By contrast, in addition to local constraints on operations and the structures they produce, the MP adds economy principles that apply either globally to sets of alternative derivations (Chomsky (1995)), or locally to sets of possible operations at each point in a derivation (Collins (1997) and Yang (1997)). Both the foundational assumptions of the MP and the way in which these have come to be accepted by a large number of linguists previously working within GB raise a number of important issues concerning the nature of theory construction and scientific discussion in the field.

Underlying the economy principles of the MP is the idea that grammar is a *perfect* computational system for mapping a selection of lexical items (a numeration) to a pair of interfaces $\langle \text{LF}, \text{PF} \rangle$ with conceptual (semantic-pragmatic) and articulatory (phonetic) cognitive modules, respectively. The well-formedness conditions on LF and PF are externally imposed by interpretability requirements of the modules to which they provide interfaces. The syntactic component of a grammar provides an optimal derivation from lexical numerations to interface representations. Hence, economy principles ensure the efficiency of the mapping. The grammar exhibits what Chomsky refers to as *virtual conceptual necessity* in that its principles and operations provide the minimal devices required to achieve optimal derivations from lexical numerations to well-formed $\langle \text{LF}, \text{PF} \rangle$ pairs.

A significant problem with this approach is that Chomsky does not clarify the notion of perfection or optimality as a property of grammar. Attempting to render this concept precise within the MP leads to severe problems. Chomsky's claim that grammar is a perfect system cannot mean that it has a relatively low degree of computational complexity as compared to other imaginable systems of formal grammar, as this is clearly not the case. Moreover, economy principles, both global and local, add a significant degree of complexity to the MP model of grammar that local constraint-based grammars avoid.¹

Chomsky does not intend optimality to be identified with the satisfaction of a maximal number of defeasible local constraints, as in Optimality Theory. He avoids the use of ranked defeasible conditions, and he treats perfection as a property of grammar rather than a relative feature of a derivation.

¹See Johnson and Lappin (1997) and (1999) for detailed discussion of this point, as well as extensive critical treatment of some of the conceptual and empirical problems raised by the MP

Perfection can also not be identified with biological efficiency, because there is no obvious sense in which the language faculty is a more or less efficient subsystem of the human organism than any other cognitive function. If it is to be judged as optimal in a biological sense, the obvious question is in comparison to what. As no basis for such a comparison is offered, the notion of grammar as a biologically optimal system is quite meaningless.

Finally, one may suggest that the notion of perfection that Chomsky has in mind is based upon an analogy with the minima and maxima principles of physics. So, for example, air pressure in a soap bubble produces a spherical shape as the optimal geometric design for distributing this pressure. Similarly, light reflecting off a mirror takes the path of least time between two points. If this is, in fact, the sort of optimality that Chomsky has in mind, then it has no place in the theory of grammar. Minimization/maximization principles are derived from deeper physical properties of the particles (waves, vectors, etc) which satisfy them. They follow from the subatomic structure and attributes of these particles, and are not themselves basic elements of the theory. Hence they have no independent explanatory status within physics, but are reducible to other principles. By contrast, the MP takes economy conditions to be essential elements of the grammar and the optimality which they encode to be one of its defining properties.

It seems, then, that the foundational assumption of the MP rests upon an obscure metaphor rather than a precise claim with clear empirical content. In fact, the MP looks like an awkward transcendental deduction of Universal Grammar in which its adherents attempt to derive the formal properties of natural language by speculating on the set of operations and conditions that provide the shortest route from a lexical numeration to a pair of LF and PF representations. The distance metric invoked in these speculations remains undefined and ungrounded in empirical considerations.

Vagueness and imprecision are not exactly unknown in much of the history of linguistic theorizing. Taken in isolation the conceptual defects of the derivational economy notion are probably no worse in kind than earlier examples might be. What is altogether mysterious from a purely scientific point of view is the rapidity with which a substantial number of investigators, who had significant research commitments in the Government-Binding framework, have abandoned that framework and much of its conceptual inventory, virtually overnight. In its place they have adopted an approach which, as far as we can tell, is in no way superior with respect to either predictive capabilities nor explanatory power. Still more remarkable is the widespread perception that a novel approach with no more 'battle-tested results' to its credit (and far narrower cross-linguistic coverage) than its predecessor represents a conceptual breakthrough for generative grammar, advancing it to the status enjoyed by the hardest of physical sciences. Consider (as just one example) the strangely imaginative assertion by Piatelli-Palmerini in his foreword to Uriagareka's *Rhyme and Reason* that minimalist grammar

is well on its way to becoming a full-blown natural science, offering a serious promise of an advanced field of scientific inquiry whose idealizations, abstractions and deductions will eventually match in depth and subtlety those of the most advanced domains of modern science. Generative grammar is turning into a natural science already, because of what it is now, not because of what it might one day turn into ...

And Uriagareka's fictional expositor of minimalism, the Linguist, echoes this peculiar claim in his proclamation that 'this degree of elegance [of UG] seems to be extraordinary. This is what I think makes linguistics closer to physics than biology, somewhat mysteriously' (p. 60). Much of Uriagareka's book consists in invoking a variety of scientific topics that have become the staple clichés of trendy 'parascientific' chit-chat—chaos theory, turbulence, emergent macroscopic phenomena, Fibonacci sequences, the thermodynamics of living systems and so on—apparently on the assumption that these issues, so airily cited in the book, have something to do with the empirical content of the linguistic notions he advocates. This groundless aura of scientism is used to promote the view that minimalist theory has brought the study of syntax to a level of

precision and empirical coverage comparable to that of chemistry and physics. In fact, Uriagareka does not manage to specify even a remotely credible connection between the concepts and methodology of the MP and those of the natural sciences. Clearly a significant number of former GB adherents must share Uriagareka's view. Why else would they simply discard a theoretical framework which they had previously thought to be the best one available for achieving genuine insight into the nature of human linguistic ability?

A comparison with paradigm change, at the level of the individual researcher, in the more established sciences may be instructive. Arthur Compton won the 1927 Nobel Prize in physics for his quantum-theoretic explanation of wavelength shifts in high-frequency X-ray/electron scattering. He was not a devotee of quantum physics when he began his research on the problem, and he used his considerable technical gifts to defend classical electrodynamics by constructing an account based on a contemporary version of Maxwell's theory of electromagnetism. It was the irreducible discrepancy between his classically predicted results and the observed angle-dependent frequency spectrum of the output radiation that led him to adopt a quantum mechanical account, which yielded a precise and accurate prediction of the observed spectrum. Only this dramatic experimental evidence for quantum physics caused Compton to give up the classical view. A generation earlier, physicists had adopted special relativity because of its success in predicting experimental and observational outcomes that classical mechanics failed to explain. In the following generation they came to embrace quantum field theory because of its empirical superiority to standard quantum mechanics in accounting for phenomena such as the Lamb shift in the hydrogen atom energy spectrum. In none of these cases did scientists abandon the highly successful models that they and their predecessors had developed except under pressure of compelling *scientific* motivation. This consisted (primarily) in the emergence of a new formalism that preserved the results of the previous paradigm while contributing an empirically grounded model of a phenomenon that remained unexplained—and *in principle* unexplainable—on the earlier account.

If linguists wish to use the practices followed in the natural sciences as a guide, then it would be reasonable to expect the catalyst for the transition from GB to the MP to be a significant body of results that follow directly from minimalist principles, but are unavailable on any plausible version of GB theory. But we see nothing of the kind in the comparison between the MP and earlier avatars of transformational grammar. Why, then, are we witness to a mass rejection of the previous decade and a half of linguistic theory? Even the most cursory examination of the way in which physics conducts its daily business and of the results that it has achieved is sufficient to expose Piatelli-Palmerini's and Uriagareka's claims as radically misguided. The MP simply does not bear serious comparison, for either accuracy or elegance, to Lagrange's eighteenth century formulation of classical mechanics, let alone to the sophisticated contemporary subdisciplines of the physical and mathematical sciences that *Rhyme and Reason* invokes, mantra-like, to such little effect.

The ease and speed with which so many GB theorists have discarded the theoretical framework in which they had invested so much research effort and embraced the bizarrely vague and unmotivated assumptions of the MP thus suggest that in large sections of the field theoretical commitment has little to do with evidence or argument. It is entirely reasonable for Chomsky to pursue his own research program on the basis of his intuition that it provides an interesting and potentially fruitful line of inquiry. Clearly, the burden of evidence and persuasion is on him to show us that this program is worth taking up. What is not readily comprehensible is that large numbers of researchers should substitute one theory for another simply on the basis of Chomsky's personal authority, without subjecting his assumptions to the sort of critical evaluation that they would normally apply to theoretical innovations proposed under different authorship.

Again, comparison with the hard sciences is instructive. Einstein, perhaps the greatest physicist ever, and Hilbert, one of the seminal figures in the history of mathematics, were both profoundly mistaken about the most important components of their respective mature scientific programs. Einstein's efforts to formulate a unified theory of the gravitational and electromagnetic field was undermined not only by his omission of nuclear forces, but by his effort to extend the Riemannian geometric formalism of general relativity to embrace the electromagnetic field. While continuing to venerate him the rest of the physics community moved in an entirely different direction, one that had far greater empirical success. Hilbert's formalist program for

mathematics consisted in an effort to reduce truth to provability. Kurt Gödel showed this program to be in principle untenable in 1930, when he proved the incompleteness of any mathematical system with sufficient expressive power to encode elementary number theory. He was an obscure mathematician at the time, almost unknown outside of Vienna. The difference in reputation between Hilbert and Gödel did not prevent the complete collapse of the formalist program. In fact, within a decade Hilbert, in a book co-authored with Paul Bernays, constructed an original proof of Gödel's incompleteness theorem, showing that he fully appreciated the implications of Gödel's work for his own program.

These incidents in the history of science show that the very greatest of minds can be deeply mistaken. More importantly, they indicate clearly that in real science, arguments from authority do not, in general, determine the direction in which a field develops. There is no sense in which the scientific status of the GB framework approaches that of physics at any period (or even, for that matter, that of contemporary geology or biology). However, the equanimity with which so many of its adherents abandoned it primarily on the strength of Chomsky's personal speculations, in the absence of any compelling scientific motivation, is a disturbing reflection of the fact that the constraints of evidence and rational argument appear to play a strikingly tenuous role in generative grammar's transformationalist branch.

In the introduction to Chapter 4, Chomsky (1995) says

It is, furthermore, far from obvious that language should have anything at all like the character postulated in the Minimalist Program, which is just that: a research program concerned with filling the gaps and determining the answers to the basic questions raised in the opening paragraph of the introduction, in particular, the question "How 'perfect' is language?"

Just so. Given the lack of any obvious content in the basic question that the MP takes as its starting point, it is particularly surprising that so many linguists have chosen to adopt its assumptions without demanding additional clarification or empirical motivation. We know of no serious scientific discipline where theoretical paradigms are granted large scale acceptance in such a cavalier and uncritical manner. It should come as no surprise, then, that linguists who exhibit so little respect for the basic theoretical assumptions that guide their own work should be relatively immune from interest in work done in alternative theoretical paradigms. We seem forced to the conclusion that a not insignificant part of our field is labouring under the manufacture of consent.

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